COMP-2650 Computer Architecture I: Digital Design Winter 2026

Lab Exercise 6 – Sequential Circuits

Due: Friday, 13 March, 2026

- *Draw the input and output lines neatly and carefully as indicated during the lectures.*
- *Refer textbook and lecture slides for more details on the parts below. Ask questions if you experience any difficulties.*

Question 1. [3+2+5=10 marks]

a) Draw the complete circuit diagram for a SR-flip flop using <u>only</u> NAND gates.
b) Provide the truth table for the circuit in part a)
c) Which state in SR latch is not stable? Briefly explain the reason using a tracing diagram of SR latch for respective state.

Question 2. [10 marks]

a) Draw the complete circuit diagram for a ***JK flip flop***, using fundamental logic gates. Label your diagram.
b) Design and draw an equivalent circuit for a ***D-flip flop*** using <u>only</u> a ***JK-flip flop*** and NOR gate(s). Use a block diagram for the ***JK-flip flop***.

Hint: Provide complete Gate Implementation, no block diagram in the implementation.

Question 3. [10 marks]

a) Show the T flip flop using only NAND gates. *Note: Understand why JK flip flop is called as Universal flip flop.*
b) Demonstrate sequential enabling in terms of cascaded two SR circuits.

Question 4. [10 marks]

a) Flip flop Excitation tables has the transition states from present to next state, Determine the inputs for D and T flipflop based on the given transitions in the states.
b) Can a latch be sub-component of a flipflop? Explain how?

Question 5. [10 marks]

Draw a block diagram (not a gate-level diagram) of a D latch and a D flip-flop. Show and label all inputs and outputs. Write the truth tables for a D latch and mention states in the truth table.
Note the difference in the latch and flip flop

FINAL NOTES:

Submission:

A. Submit the lab work on Brightspace before the deadline.
B. Students are responsible for uploading the work in high resolution when submitted as images.
C. No extensions will be granted. Students are provided five days to complete the assignment, and it is student's responsibility to manage the time and complete the assignment before the submission deadline.

Evaluation:

A. Students are evaluated on all stated requirements.
B. It is mandatory that students complete their work and must be able to justify their answers when asked to do so by instructors and teaching staff.
C. If the work is not clearly written or presented, or submission is in unsupported file type, it will be graded as zero.